

Sultant Of Oman
Ministry of Health
Royal Hospital



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Sultanate of Oman
Ministry of Health
The Royal Hospital
Department of Blood Transfusion Services

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Title: Transfusion in patients with Thalassaemia

Introduction:

These patients are usually requiring regular red cell transfusion and they are at risk of transfusion related adverse events including red cell allo-immunization, febrile non haemolytic transfusion reaction, haemolytic transfusion reactions (both immediate and delayed). Therefore, it is prudent to try and minimize the above reactions through the use of appropriate type of red cell units (leucodepleted red cells), strict record keeping (history of allo-antibodies within our institution or if known in other institutions, baseline red cells phenotyping).

Scope:

This document is applicable to adult patients with both thalassaemia major and intermedia.

Pre-first transfusion:

All newly diagnosed thalassemia patients should have their red cell phenotype done at diagnosis prior to any transfusion or last history of transfusion was more than past three months.

Red cells units' specifications:

- Thalassaemia patients preferably receive leucodepleted red cells units (Currently the available leucodepleted red cells units are suspended in SAGM)
- Preferably less than 2 weeks old from donation.
- Sickle cell negative.

Request of red cell units:

- The pre-transfusion hemoglobin levels should be calculated using the patient's weight and the pre-transfusion hemoglobin level and the required volume of blood transfused in each visit(Refer to table-1)

Table-1:

Parameters to be checked	CBC and cross-match samples taken			
Pre-transfusion Hb (gm/dl)	Hb>9	Hb 8-9	Hb 7-8	Hb <7
Target Hb	Raise Hb to 14gm/dl [Max 800-900 ml]	Raise Hb to 13gm/dl [Max 800ml]	Raise Hb to 12gm/dl [Max 600 ml]	Raise Hb to 10gm/dl [Max 400 ml]
Next appointment	4 weeks	3 weeks	2 weeks	1 week
<i>Required PRBC's volume in ml = (Target Hb – Pre-transfusion HB) X patient weight X 4</i>				
<i>[Maximum 20ml/Kg or 800-900 ml PRBC per transfusion visit (whichever quantity is less)]</i>				

Blood samples:

- All samples collected from daycare ward should be sent to the hematology lab in red labeled bags with "thalassemia" written on them.
- Once received by hematology staff, samples will be processed immediately.
- Hemoglobin results should be released with-in 30 minutes of receiving the samples. (CBC. TAT= 30min)
- The blood bank staff receiving the cross-match sample should be able to view the hemoglobin and weight of the patient on the transfusion request screen.
- Once Hb level is resulted, the required volume is to be calculated by the assigned physician in the daycare
- Antibody screen (ANT) will be done in every visit and should be archived.
- Antibody Investigation "identification" (ANID) should be performed when required and results should be archived.
- Patient with known alloantibody/s or Rh (D) negative blood group should have their cross-match sample sent few days prior to their transfusion visit to facilitate the availability of suitable units.
- Compatible units should be ready for transfusion within 2 hours after sample registration (Exemption new cases/ known case of antibodies).
- The volume of each blood unit should be written on the cross-match label.
- The assigned blood bank staff should also write the final cumulative reserved unit/s volume/s for each patient on the issuing form. This information is very important to decide on the next transfusion visit appointment.

blood bank.

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Review Details

	Review BY	Review Date
1	Dr Sabriya Al Hashami	24/01/2024

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